Lookup sheet: Pták–Pulmannová, *Orthomodular Structures as Quantum Logics* (1991)

Printed 2026-11-10. Reviewed: false. Chapter locations unverified: use contents and index.

Our case, to hold beside the book. $\Omega = \mathbb{N}^3$. Three σ-algebras, one per pair of coordinates: events depending only on (x_0, x_1), on (x_1, x_2), on (x_2, x_0). D_3 = the smallest collection containing all three and closed under complement and countable disjoint union. A *state* is a finitely additive $s : D_3 \to \{0, 1\}$.

Words (ours → likely theirs). carrier D_3 → concrete (σ-)logic, σ-class; windows → blocks, Boolean sub-σ-algebras; 0/1 state → two-valued state / measure; given by a point → concentrated, carried by a point.

Index words to try: *concrete logic, σ-class, set-representable, two-valued, concentrated, pasting, generated, extension, countable, determined, unital / rich / full set of states.*

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1. Normal form (the one that bears on the open mathematics).
A description of the members of the σ-class *generated by* two or three sub-σ-algebras, e.g. "every member is a disjoint union of sets, one from each". We have this for one round of patching only.

No. _____ **p.** _____ ☐ found ☐ not there (sections covered: _____)
Hypotheses, word for word:

1′. Determination. Any theorem: "a state on a concrete logic is determined by its restrictions to the generating blocks / sub-σ-algebras", or a counterexample to it.

No. _____ **p.** _____ ☐ found ☐ not there (sections covered: _____)
Statement:

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2. A two-valued state given by no point, on pairwise-compatible blocks.
An example of a concrete logic, ideally with three blocks where each pair lies in a common Boolean algebra, carrying a two-valued state that is not concentrated at a point and not the restriction of an ultrafilter of the underlying set. (Already on our shelf: Navara–Pták 1983. Note whether the book only repeats it.)

No. _____ **p.** _____ ☐ found ☐ not there (sections covered: _____)
The example (set, blocks, state):

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3. Countable underlying set.
A two-valued state on a concrete σ-class of subsets of a *countable* set that has no σ-additive extension, or any theorem saying this cannot happen.

No. _____ **p.** _____ ☐ found ☐ not there (sections covered: _____)
Statement:

Skip: Hilbert-space logics, Gleason, tensor products, anything needing lattice structure throughout.

Already known, no need to re-find: extension of a state to all subsets $\iff$ Horn–Tarski inequality (*Theory of Charges* Thm 3.1.9, 3.2.10, read 2026-11-10). That book is real-valued and has no two-valued or determination result.

What each answer does. 1 or 1′ found: possible route to, or owner of, the open step on the triangle. 2 or 3 found: the example is theirs, not ours. All four "not there": the open step stays open and the "not found" verdicts firm up.